SOCIALMIND DRIVE

AGENTIC AI FOR ROBUST AND SOCIALLY-AWARE AUTONOMOUS DRIVING



SLIIT
FACULTY OF COMPUTING

Agentic AI for Robust and Socially Aware Autonomous Driving

Proposal Comments	How the Comments Were Addressed
Focus on narrowing down the scope of the component	• C1 - The scope was narrowed to pedestrians hidden by a nearby vehicle. The model predicts whether they will Cross, Wait, or Hesitate. • C2-Focused on one controlled four-way unsignalized SUMO scenario for V2V-based decision support.
For the whole project fairness-time measurement is very important	• Added runtime fairness and performance metrics to monitor waiting, prevent starvation, and measure inference/decision time. • New verified knowledge is periodically shared with all vehicles

Agentic AI for Robust and Socially Aware Autonomous Driving

Problem, It's Impact / Gravity

- Reactive autonomous driving systems struggle with hidden human intent
- Can lead to unsafe driving, congestion, and accidents
- Limited collaboration between vehicles

Person / Group / Society / Technology Benefited

- Autonomous vehicle companies
- ADAS and smart mobility providers
- Drivers and pedestrians

Design Excellence of the Overall Solution

- Agentic AI framework for socially aware autonomous driving
- Combines intent prediction, right-of-way negotiation, Theory of Mind, and collective learning
- Vehicles collaboratively share verified knowledge for safer driving decisions
- Supports intelligent, adaptive, and socially aware autonomous behavior

Agentic AI for Robust and Socially Aware Autonomous Driving



IT22245960 | Bandara W A P S

Adaptive Intent Prediction in Occluded Urban Scenarios

Specialization: Data Science

Adaptive Intent Prediction in Occluded Urban Scenarios

1 How the Solution Addresses the Sub-Problem / Prototype

- Developed pedestrian intent prediction system for autonomous driving
- The solution predicts pedestrian behavior under partial conditions.
- Implemented a multi-class prediction model
- Proposed the OAIP model

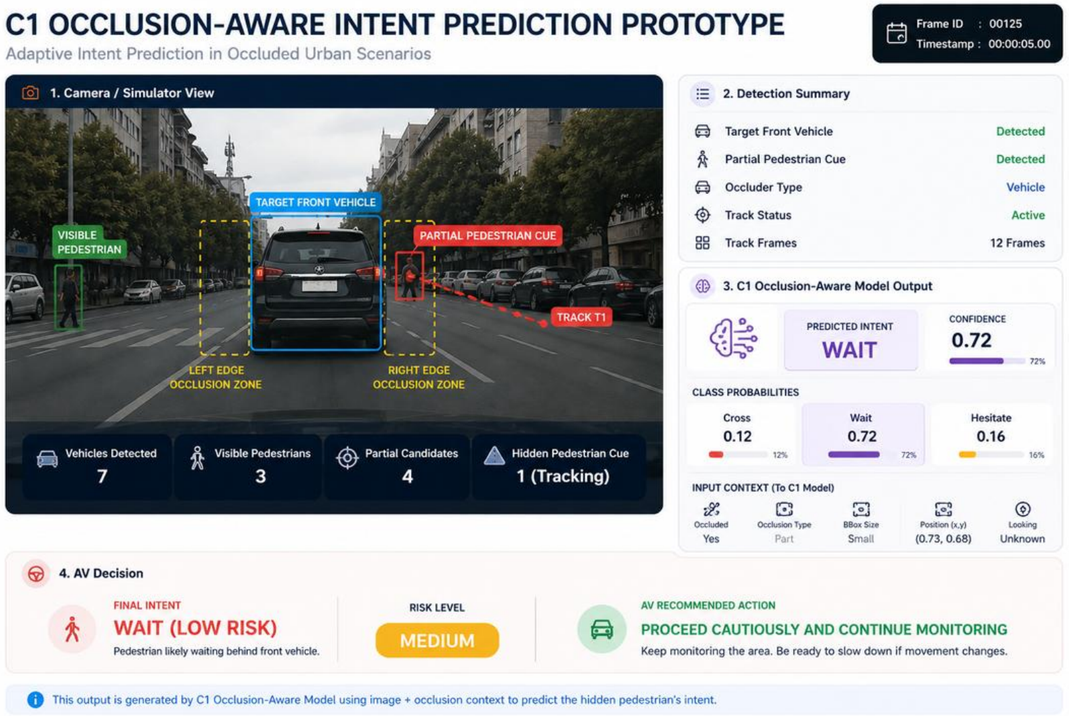
2 User Requirements Addressed by the Solution

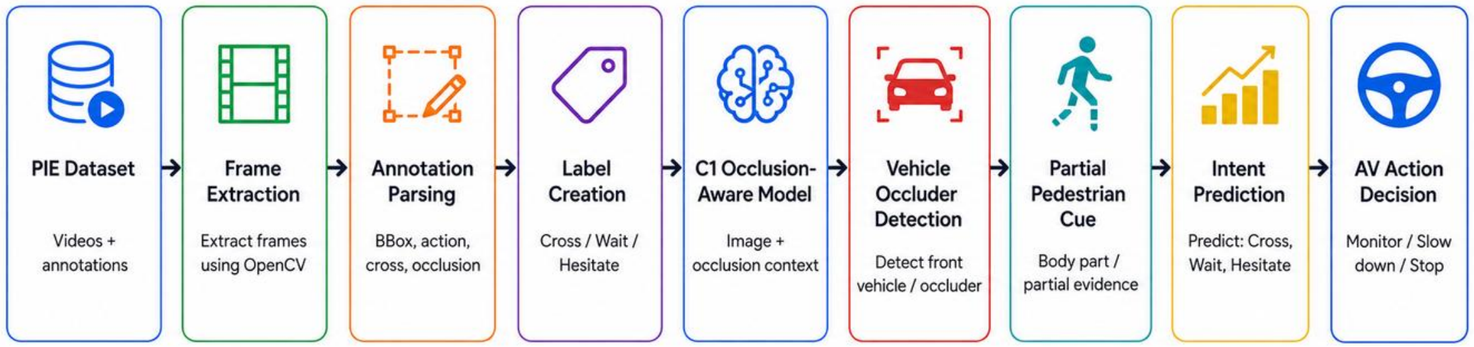
- Early prediction of pedestrian crossing intent before full visibility.
- Improved safety in urban driving environments with occluded pedestrians.
- Ability to handle complex real-world scenarios such as pedestrians behind vehicles.
- Robust performance under partial conditions.
- Real-time compatible lightweight architecture for autonomous systems.

Adaptive Intent Prediction in Occluded Urban Scenarios

3 Design Excellence / Contribution

- Developed the C1 model to predict Cross, Wait, or Hesitate.
- Added vehicle occluder detection to focus on pedestrians hidden by nearby vehicles.
- Final output supports AV decisions







IT22265906 | Nethmini R A I

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

Specialization: Data Science

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

1 How the Solution Addresses the Sub-Problem / Prototype

- Extracting vehicle movement and interaction features
- Identifying nearby and conflicting vehicles
- Predicting vehicle behaviour using LSTM
- check crossing safety
- Generating final AV action: GO / WAIT / YIELD
- Showing decisions in SUMO simulation

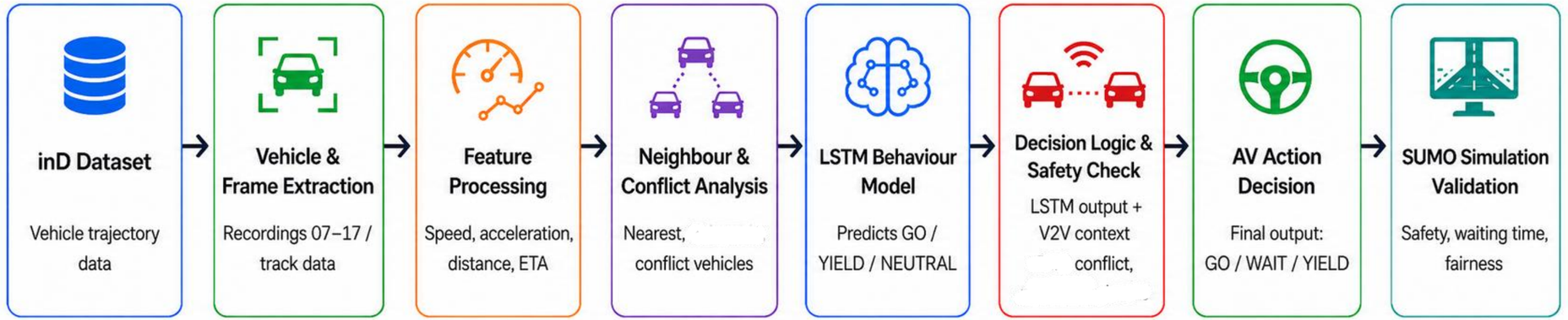
2 User Requirements Addressed by the Solution

- Use real vehicle movement data
- Create sequence data for LSTM prediction
- Classify vehicle behaviour into decision states
- Support V2V-style vehicle awareness
- Detect unsafe crossing situations
- Avoid collision during intersection crossing
- Measure fairness

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

1 Design Excellence / Contribution

- LSTM-based vehicle behaviour prediction
- V2V-inspired local awareness between vehicles
- Neighbour vehicle and conflict detection
- Proactive decision agent for intersection crossing
- SUMO prototype simulation using test scenario





IT22223012 | Withanage W W D T H

Proactive Social Compliance Modeling using Agentic Theory of Mind

Specialization: Data Science

Proactive Social Compliance Modeling using Agentic Theory of Mind

1 How the Solution Addresses the Sub-Problem / Prototype

- Extracting pedestrian movement, pose, behavior, and context features.
- Classifying the pedestrian into 6 states
- Using TTC and rule-based safety logic to decide AV action
- Showing the decision in SUMO simulation with a real PIE test sequence.

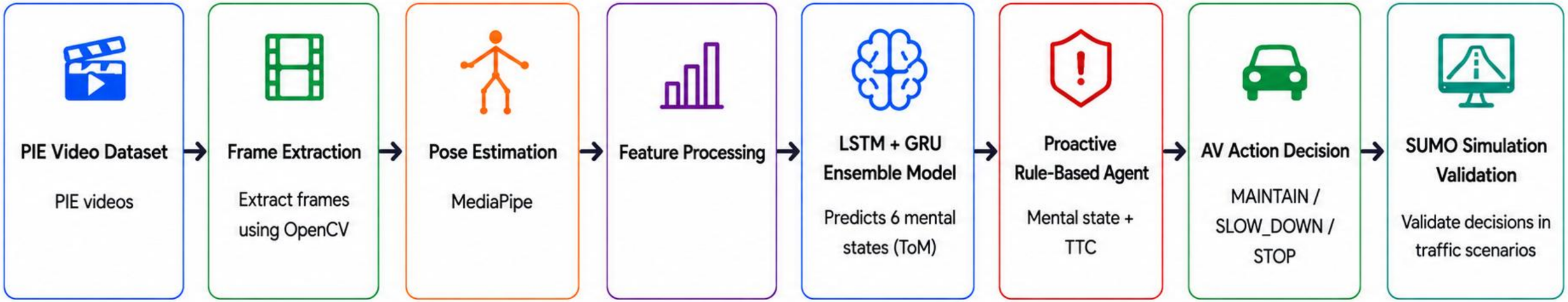
2 User Requirements Addressed by the Solution

- Extract pose landmarks
- Feature extraction
- Create 30-frame sequences
- Classify 6 mental states
- Proactively Map mental state to AV action
- Use safe fallback action when risk is high
- cultural/social behavior adaptation
- Measure inference time to ensure real-time AV decision-making

Proactive Social Compliance Modeling using Agentic Theory of Mind

1 Design Excellence / Contribution

- Theory -of-Mind inspired mental-state prediction
- Ensemble model (LSTM + GRU are combined to improve reliability)
- Proactive decision agent using rule-based reasoning.
- SUMO prototype simulation using test data





IT22022974 | Karunaarachchi K A I H

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

Specialization: Data Science

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

1 How the Solution Addresses the Sub-Problem / Prototype

- Identify known, rare, and new traffic signs
- Use an RL agent to decide SHARE or IGNORE
- Verify shared knowledge using global verification
- Broadcast verified knowledge to vehicles
- Incrementally update vehicle models using verified knowledge

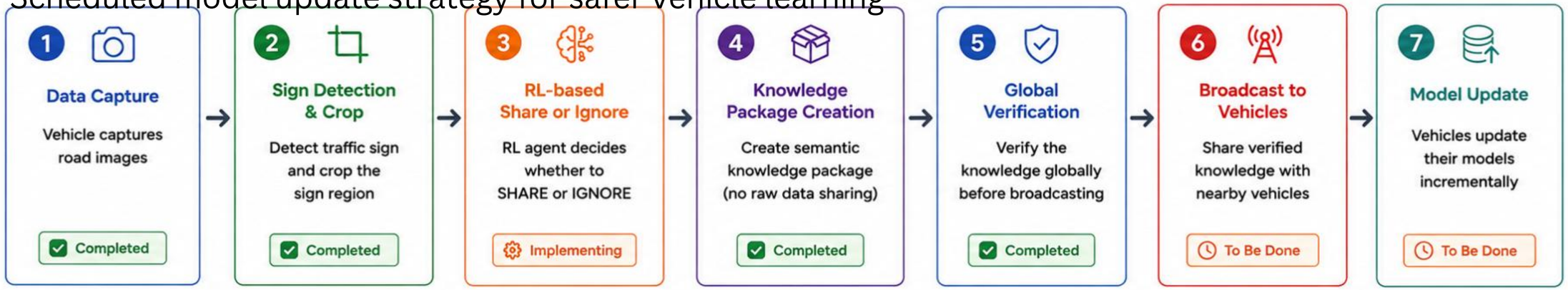
2 User Requirements Addressed by the Solution

- Successfully creates semantic knowledge packages using cropped sign data
- Reliable selective knowledge sharing mechanism
- Successfully verifies shared knowledge before broadcasting
- Supports collaborative learning between vehicles

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

1 Design Excellence / Contribution

- RL-based intelligent selective knowledge sharing mechanism
- Semantic knowledge package generation using cropped sign data
- Global verification before knowledge broadcasting
- Incremental collective learning approach for autonomous vehicles
- Scheduled model update strategy for safer vehicle learning



Agentic AI for Robust and Socially-Aware Autonomous Driving

Customer Persona

- Autonomous driving technology companies
- Smart mobility and ADAS providers
- Connected vehicle solution companies
- AI-driven transportation research organizations

Market Size

- Growing autonomous vehicle and ADAS market
- Increasing demand for safer AI-driven transportation
- Expanding smart mobility industry worldwide

What is Unique in Your Solution

- Agentic AI framework integrating pedestrian intent prediction, right-of-way negotiation, Theory of Mind reasoning, and collective vehicle learning
- Vehicles collaboratively share verified knowledge to understand hidden human behavior, handle complex road interactions, and enable safer, socially aware autonomous driving

Thank You.